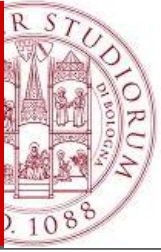




What is HCI?

Human Computer Interaction

Catia Prandi
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Hall of Fame or Shame?

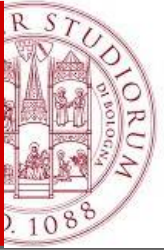




There is no good or wrong
answer: is subjective

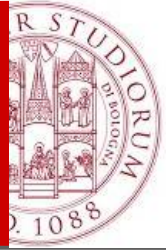
Hall of Fame or Shame?



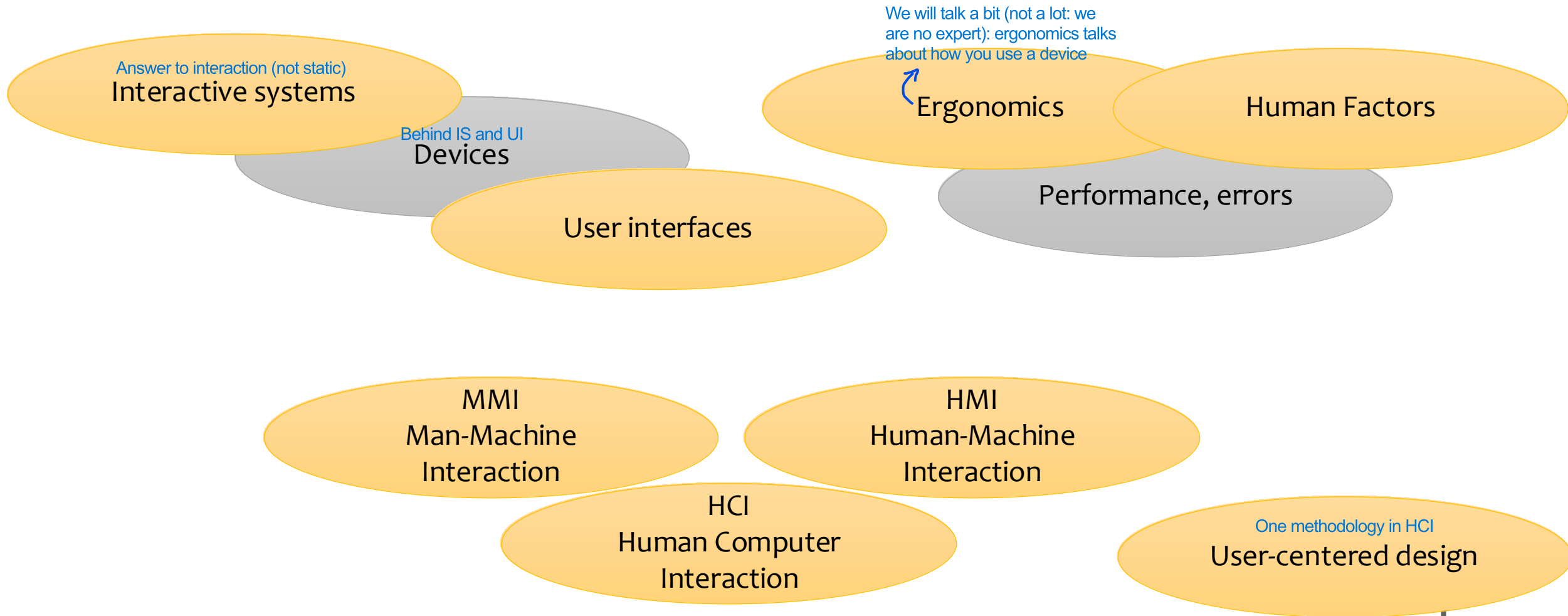


Goals

- What is HCI?
- What is usability?
- What are the Interaction Design processes, and how does they relate with Software Engineering processes?
- What is meant by User Centered Design?

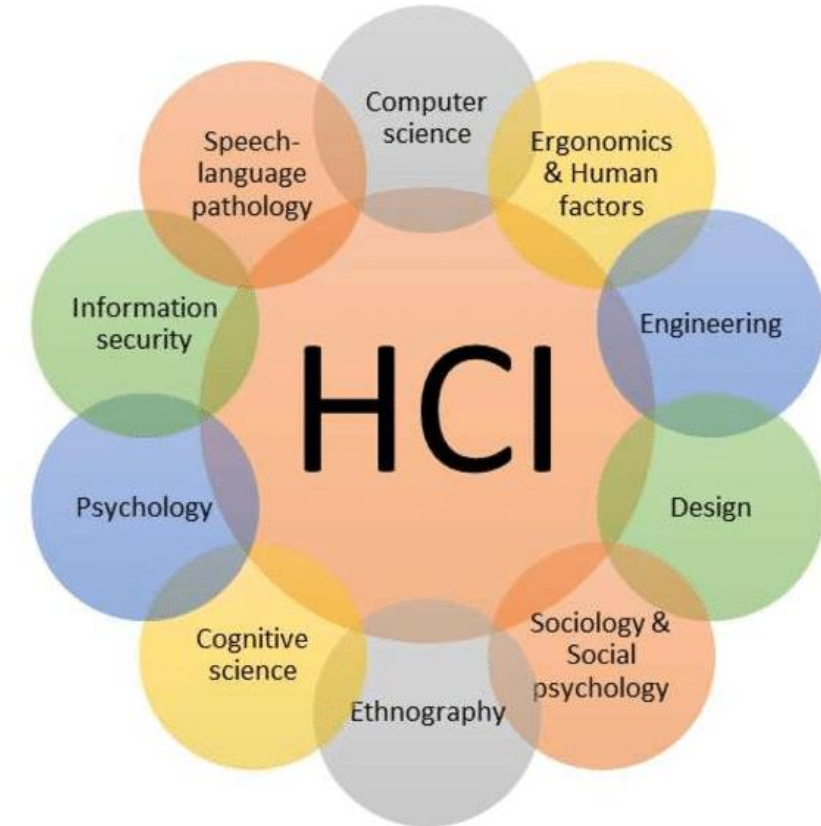
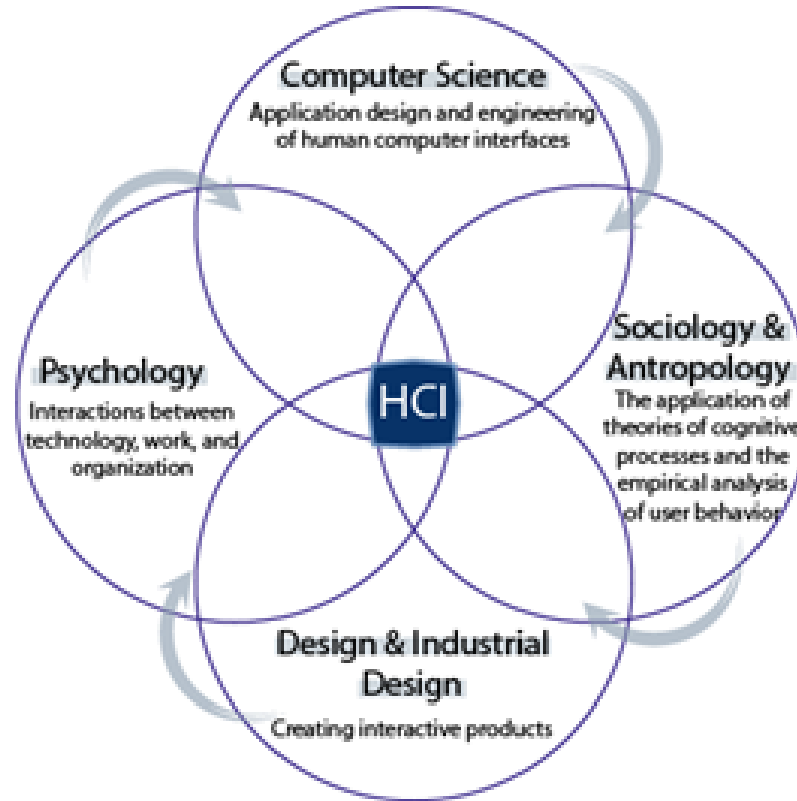


Interconnected Concepts, and Evolution



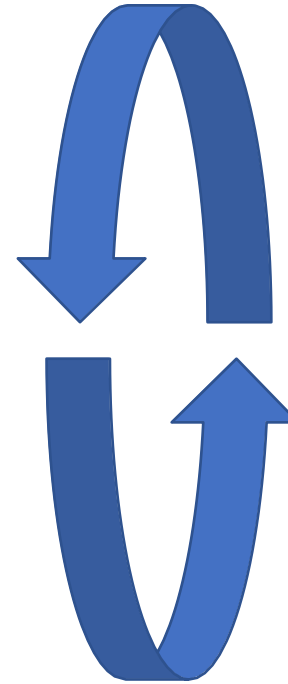
HCI: a multi-disciplinary field

Human-Computer Interaction (HCI)



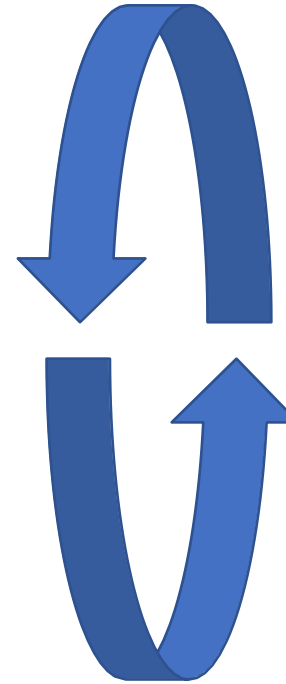
HCI Is Multidisciplinary

- Psychology and cognitive science
 - User perceptual, cognitive and problem-solving skills
- Ergonomics
 - User's physical capabilities
- Sociology
 - Understanding the wider context of the interaction
- We will focus on this **Computer Science and Computer Engineering**
 - **Building the necessary artifacts** (HW, SW)
- Business
 - Satisfying market needs
- Graphic design
 - Produce an effective interface presentation
- Technical writing
 - Documentation, manuals, on-screen content
- ...



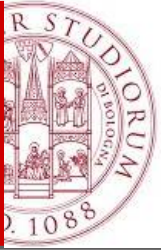
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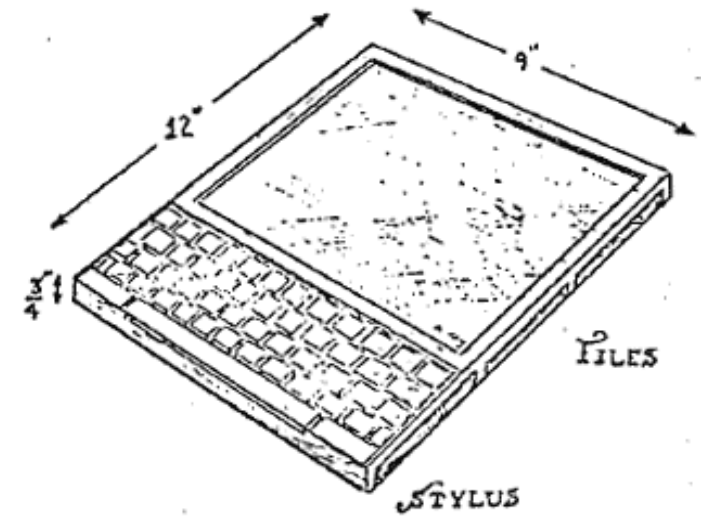
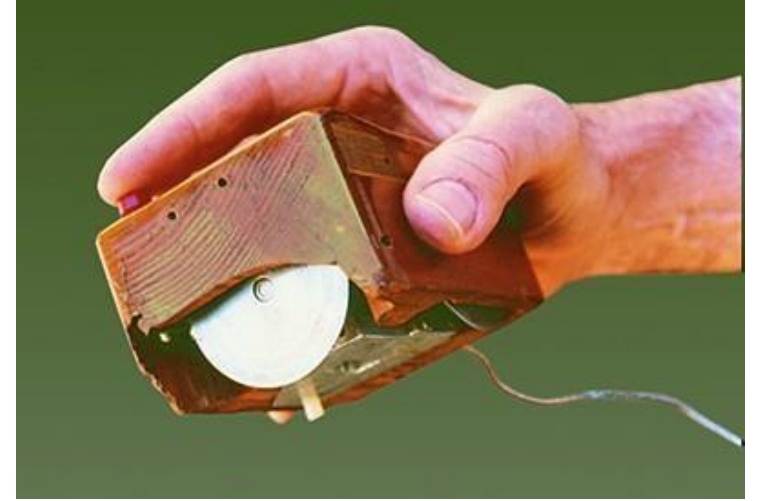
To help us in applying expertise from many different fields:

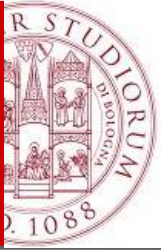
- Design methods and processes
 - Models
 - Heuristics
 - Best practices
 - Conventions
 - Experiments and user studies
- to know if our design meet the initial needfinding



Human-Computer Interaction... In Brief

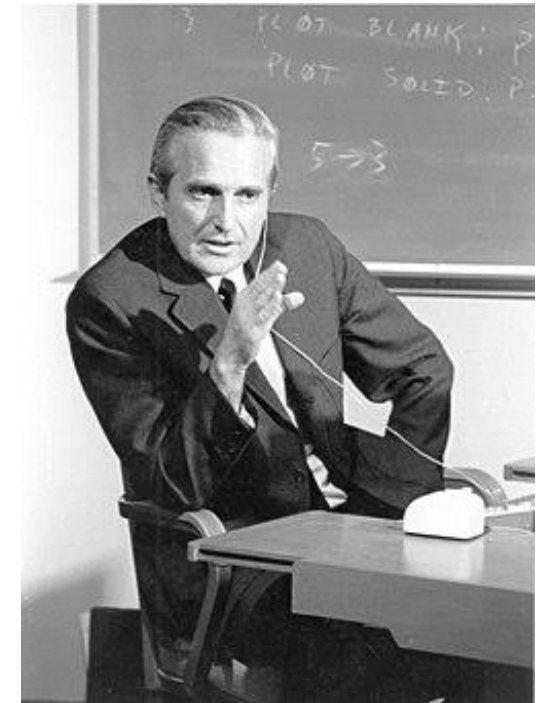
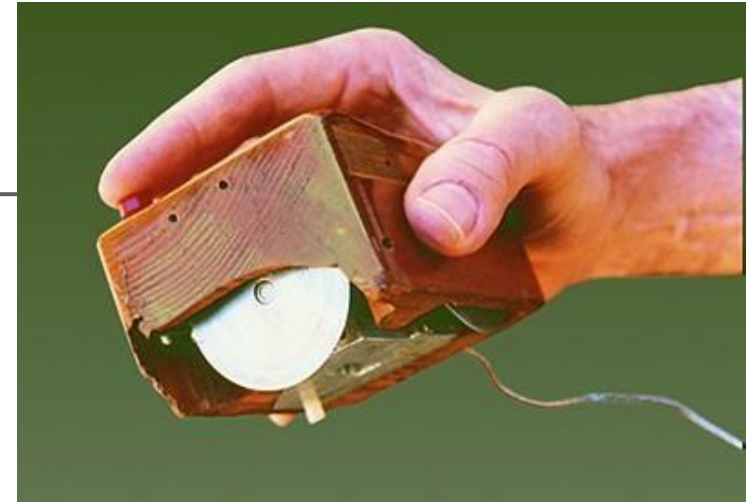
- A multi-disciplinary field
- Concerned with the **design, evaluation, and implementation** of interactive computing systems for human use
 - and with the study of **major phenomena** surrounding them





The first mouse prototype

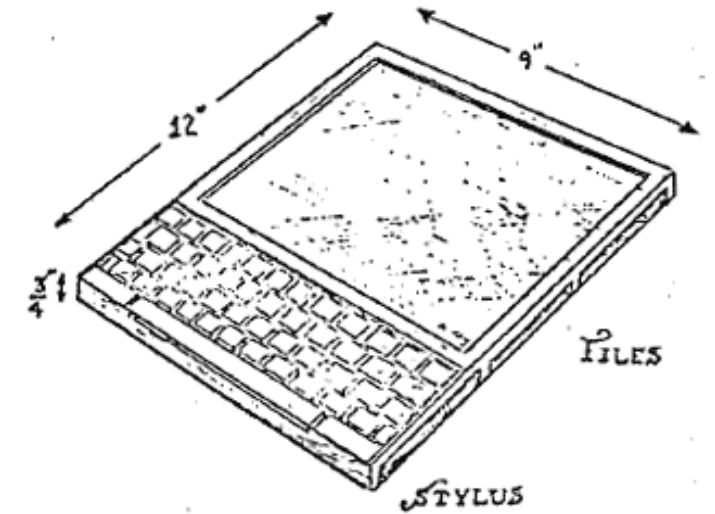
- In 1964, inventor [Douglas Engelbart](#) holding the first computer mouse, showing the wheels that make contact with the working surface
- On 9 December 1968, Engelbart publicly demonstrated the mouse at what would come to be known as [The Mother of All Demos](#)



He is best known for his work on founding the field of human-computer interaction, particularly while at his Augmentation Research Center Lab in SRI International, which resulted in creation of the computer mouse, and the development of hypertext, networked computers, and precursors to graphical user interfaces.

The Dynabook prototype

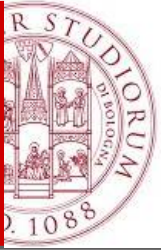
- Envisioned by [Alan Kay](#) in 1968 while a PhD candidate, and later developed and described as the **Dynabook** in his 1972 proposal "A personal computer for children of all ages"
- Never became a product..
- (Microsoft came up with its tablet PC in 2001..)



Alan Kay pioneered work on **object-oriented programming** and **windowing graphical user interface (GUI)** design. At Xerox PARC, he led the design and development of the first modern windowed computer desktop interface. There he also led the development of the influential object-oriented programming language Smalltalk, both personally designing most of the early versions of the language and coining the term "object-oriented."

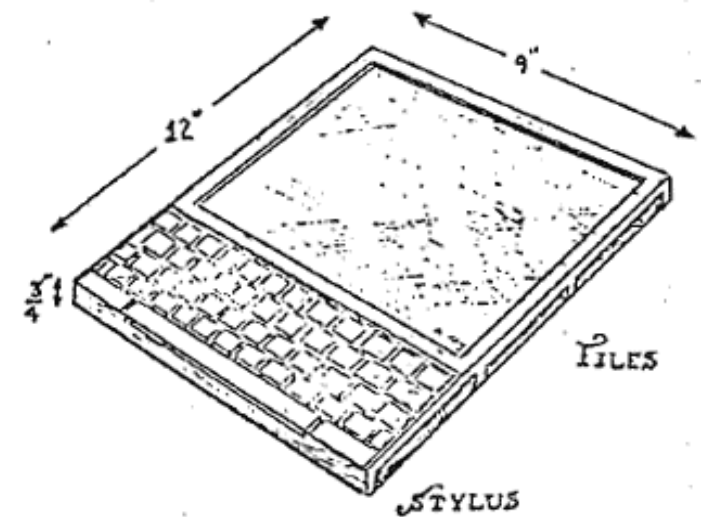
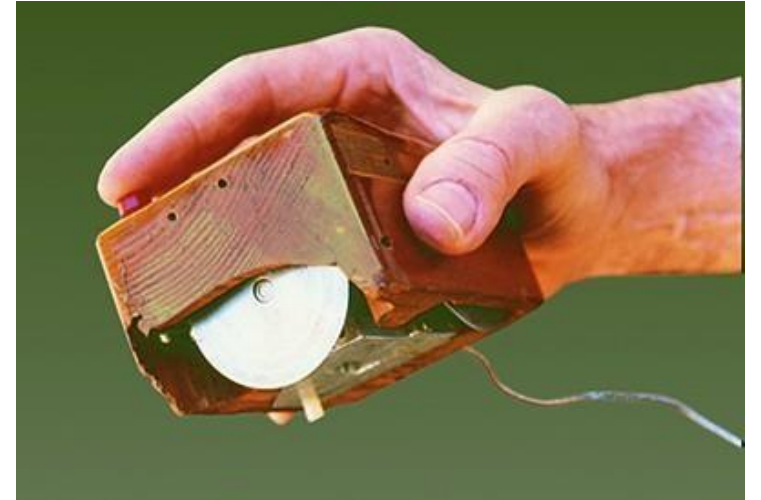
<https://mprove.de/visionreality/media/kay72.html>

<https://mprove.de/visionreality/media/Kay72a.pdf>



Human-Computer Interaction... In Brief

- A multi-disciplinary field
- Concerned with the **design, evaluation, and implementation** of interactive computing systems for human use
 - and with the study of **major phenomena** surrounding them
- Involves two **entities** (the human and the computer) that **determine** each other behavior over time
 - framed in terms of **humans' goals** and **related tasks/pursuits**



- What goals and tasks are?

Example:

User Goal: enroll in HCI exam

User Tasks:

- Go to alma esami
- Login
- Enroll in HCI
- ...
- ...



The Goal of HCI

Define the user's tasks
with a focus on Usability

Ingredients

We need to think about:

- The User(s)
- The Computer(s)
- The Task(s) to be accomplished

L'interazione è il modo in cui gestiamo e supportiamo il task dell'utente. Viene in un secondo momento nel processo di progettazione: prima definiamo il task dalla prospettiva umana e solo successivamente traduciamo tale task in interazioni specifiche.

Goal

- The system **MUST** support the user's **task**, with a focus on its **usability**

- Useful
- Usable
- Used

Characteristics
of Usability

Useful: Il sistema offre le funzionalità necessarie per permettere all'utente di raggiungere i propri obiettivi.

Usable: Il sistema permette all'utente di svolgere i propri task in modo semplice, efficace, efficiente e soddisfacente.

Used: È il risultato finale e concreto.

The Ingredients

The human

When we talk about Human, we need to consider all of these aspects

■ Sensory systems

- Visual
- Auditory
- Haptic
- Spatial
- (others?)

indica i canali attraverso cui l'essere umano percepisce le informazioni e gli stimoli provenienti dal computer o dall'ambiente circostante.

■ Acting systems

- Hands
- Voice
- Head, Body, ...

indicano le parti del corpo e i meccanismi fisici con cui l'utente agisce sul computer per inviare comandi o manipolare l'interfaccia.

■ Cognitive processes

- Perception
- Memory

The computer

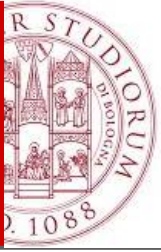
Peripherals to help the user to complete the tasks and to reach the Goal (interact with computer)

■ Input peripherals

- Keyboard, mouse
- Trackpad, trackball
- Touch surfaces or screens
- Microphone
- Sensors
- Card readers
- ...

■ Output peripherals

- Screen
- Audio (voice, sounds)
- Haptics
- VR/AR headsets
- ...



The Ingredients

The human

- Sensory systems
 - Visual
 - Auditory
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- Acting systems
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 - ...

Sensing the future of HCI: touch, taste, and smell user interfaces

Authors:  Marianna Obrist,  Carlos Velasco,  Chi Vi,  Nimesha Ranasinghe,  Ali Israr,  Adrian Cheek,  Charles Spence,  Ponnampalam Gopalakrishnakone | [Authors Info & Claims](#)

Interactions, Volume 23, Issue 5 • Pages 40 - 49 • <https://doi.org/10.1145/2973568>

Published: 23 August 2016

76  18,170

IMAGE BY INSAGO / SHUTTERSTOCK.COM

Insights

- Touch, taste, and smell are an underexplored design space in the context of interactive experiences.
- The integration of the physical and chemical senses into technology will transform existing interaction paradigms.
- Multisensory research creates new knowledge and stimulates the appetite for innovation.

The senses we call upon when interacting with technology are restricted. We mostly rely on vision and hearing, and increasingly touch, but taste and smell remain largely unused. Although our knowledge about sensory systems and devices has grown rapidly over the past few decades, there is still an unmet challenge in understanding people's multisensory experiences in HCI. The goal is that by understanding the ways in which our senses process information and how they relate to one another, it will be possible to create richer experiences for human-technology interactions.

To meet this challenge, we need specific actions within the HCI community. First, we must determine which tactile, gustatory, and olfactory experiences we can design for, and how to meaningfully stimulate them in technology interactions. Second, we need to build on previous frameworks for multisensory design while also creating new ones. Third, we need to design interfaces that allow the stimulation of unexplored sensory inputs (e.g., digital smell), as well as interfaces that take into account the relationships between the senses (e.g., integration of taste and smell into flavor). Finally, it is vital to

Models of Interaction

A general framework to understand how User and System interact



First we understand Human needs, then computational part and then interaction (a way to answer the user task)

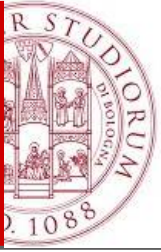
What Is “Interaction” (in HCI)?

■ Interaction is...

There are different way to interact with the system (no need to study the table)

Concept	View of interaction	Key phenomena and constructs	Good interaction	Example support for evaluation and design
Dialogue	a cyclic process of communication acts and their interpretations	mappings between UI and intentions; feedback from the UI; turn taking	understandable; simple, natural; direct	methods/concepts for guessability, feedback, mapping; walkthroughs
Transmission	a sender sending a message over a noisy channel	messages (bits); sender and receiver; noisy channels	maximum throughput of information	metrics and models of user performance
Tool use	a human that uses tools to manipulate and act in the world	mediation by tools; directness of acting in the world; activity as a unit of analysis	useful and transparent tools; amplification of human capabilities	compatibility in instrumental interaction; break down analysis
Optimal behavior	adapting behavior to goals, task, UI, and capabilities	rationality; constraints; preferences; utility; strategies	improves or reaches maximum or satisfactory utility	models of choice, foraging, and adaptation
Embodiment	acting and being in situations of a material and social world	intentionality; context; coupling	provides resources for and supports fluent participation in the world	studies in the wild; thick description
Experience	an ongoing stream of expectations, feelings, memories	non-utilitarian quality; expectations; emotion	satisfies psychological needs; motivating	metrics of user experience; experience design methods
Control	interactive minimization of error against some reference	feedforward; feedback; reference; system; dynamics	rapid and stable convergence to target state	executable simulations of interactive control tasks

Taken from: Kasper Hornbæk & Antti Oulasvirta, What Is Interaction? In: *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*



What Is Interaction (in HCI)?

■ Interaction...

- is not the idea promoted and repeated in folk notions that a computer and a human are engaged
interagendo
L'interazione non è una semplice conversazione statica o una risposta automatica del tipo "io premo un tasto, il computer mostra qualcosa". Vederla solo come uno scambio di messaggi banale è riduttivo.

- it concerns two entities – humans and computers – that determine each other's behavior over time
L'interazione è un processo dinamico continuo in cui l'umano e il computer si influenzano e si modificano a vicenda a ogni passaggio. Le azioni dell'utente cambiano lo stato del sistema, e le risposte del sistema guidano o vincolano la decisione successiva dell'utente.

- Their mutual determination can be of many types, including statistical, mechanical, and structural

■ Users, with their goals and ^{tasks} pursuits, are the ultimate metric of interaction

→ We translate the Goals in something that is possible to interact with (interactions)



When you need to define the goals, it is important to define the Domain. Then, for each Goal, we need to define the tasks to reach the Goal

Assumptions

The system must speak the same language of the user using it (so define the domain and the target user)

- The **user** wants to accomplish some **goals**, in a specific application **domain**
 - Each domain has a specific jargon/slang, set of possible processes and goals, artifacts and building blocks, ...
- **Tasks** are operations to manipulate the concepts of a domain
 - The goal ^{viene raggiunto} ~~is attained~~ by performing one or more tasks
- Interaction studies the relation between User and System
 - The system possesses a **state** and “speaks” a **core language**
 - The user possesses a **state**, that includes an **understanding** of the system’s state, some **intention** to perform a task, and “speaks” the **task language**

Task Language vs Core Language: Il Task Language (linguaggio del dominio/utente) è la terminologia usata dagli utenti per esprimere i propri obiettivi (es. "isciversi a un esame", "compilare un file"). Il Core Language (linguaggio del sistema) è lo stato interno e la logica con cui il computer elabora i dati.

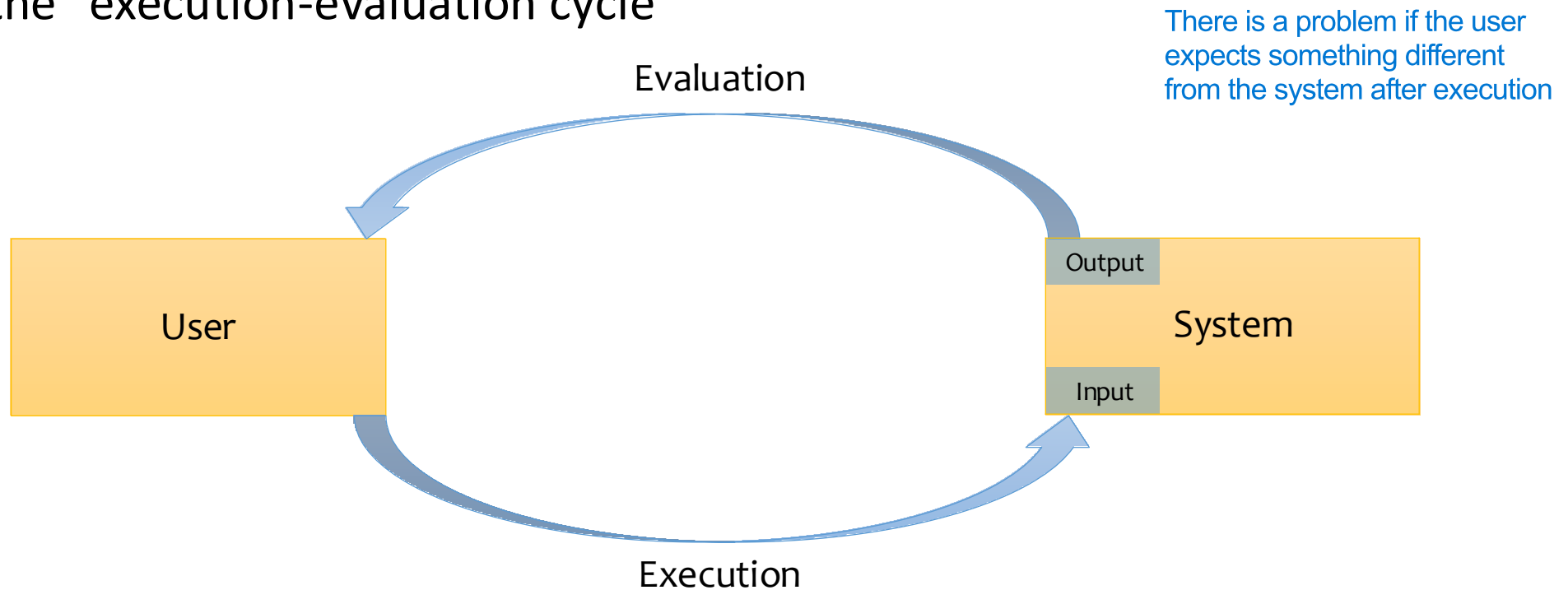
L'obiettivo dell'interfaccia (UI Language): Il ruolo dell'interfaccia utente è proprio fare da traduttore tra il Task Language dell'utente e il Core Language del sistema (fasi di articulation e presentation).

Definizione dei Task: Quando si definiscono i task, ci si posiziona sempre dalla prospettiva dell'umano e del suo dominio applicativo, non da quella interna del computer.

When defining Goals and related Tasks, we need to understand the domain language (or task language). For example, if the system is designed for CS students, we must adopt their domain terminology to define their goals and tasks. The system language (core language), on the other hand, is the internal language used by the system to process those operations.

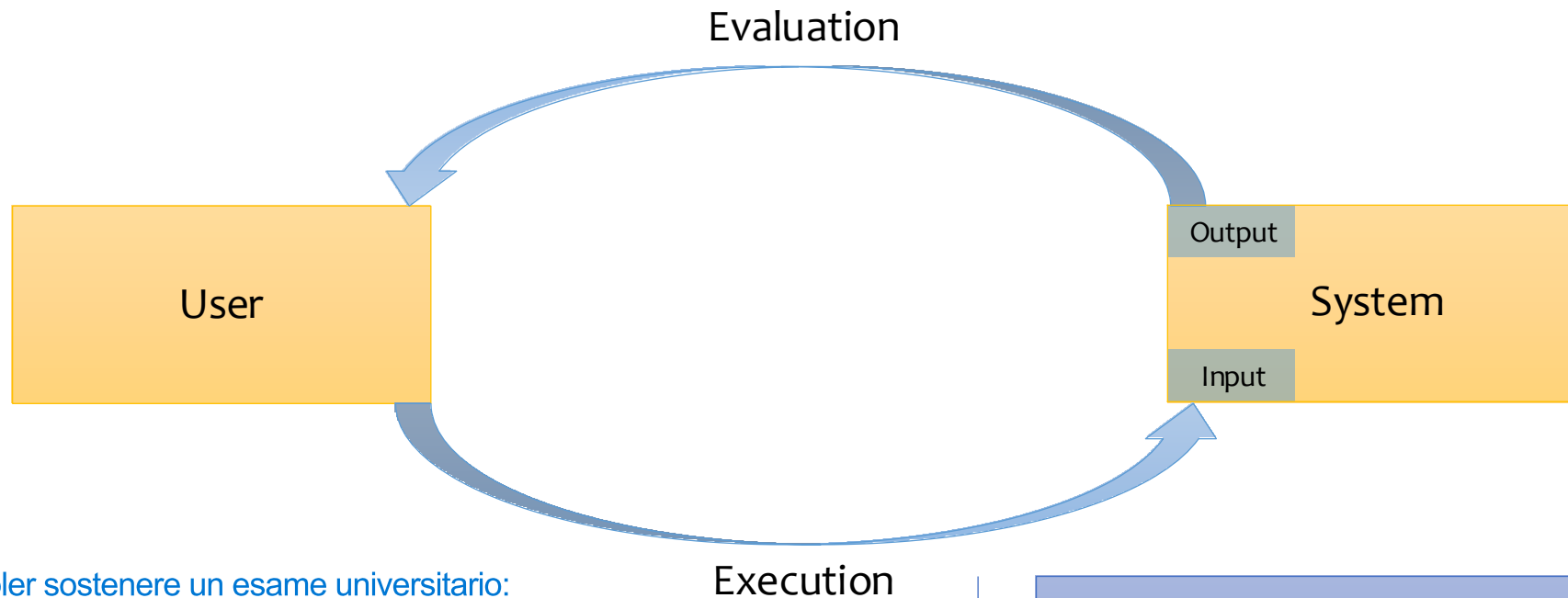
Norman's Model of Interaction

Also called the “execution-evaluation cycle”



The user has Goals and performs a sequence of Tasks (actions) to reach them. The user sends inputs to the system (Execution) and receives outputs to check the results of those actions (Evaluation). The Norman's model is simple and focusses on Human and Computer instead of interactions

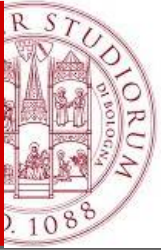
Norman's Model of Interaction



Esempio: Immagina di voler sostenere un esame universitario:

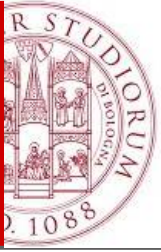
- Establishing the goal (Obiettivo): "Voglio iscrivermi all'appello di HCI".
- Forming the intention (Intenzione): Decidi che il mezzo per farlo è accedere al portale universitario (AlmaEsami) e selezionare l'esame dalla lista delle iscrizioni aperte.
- Specifying the action sequence (tasks sequence per arrivare al Goal): Pianifichi la sequenza fisica dei clic: aprire il browser, digitare l'URL, fare clic sul pulsante "Login", inserire la password, fare clic su "Iscriviti".
- Executing the action (Esecuzione): Muovi le dita sulla tastiera e il mouse per fare i clic previsti.

1. Establishing the goal - WHAT
2. Forming the intention - HOW
3. Specifying the action sequence ^{tasks}
4. Executing the action

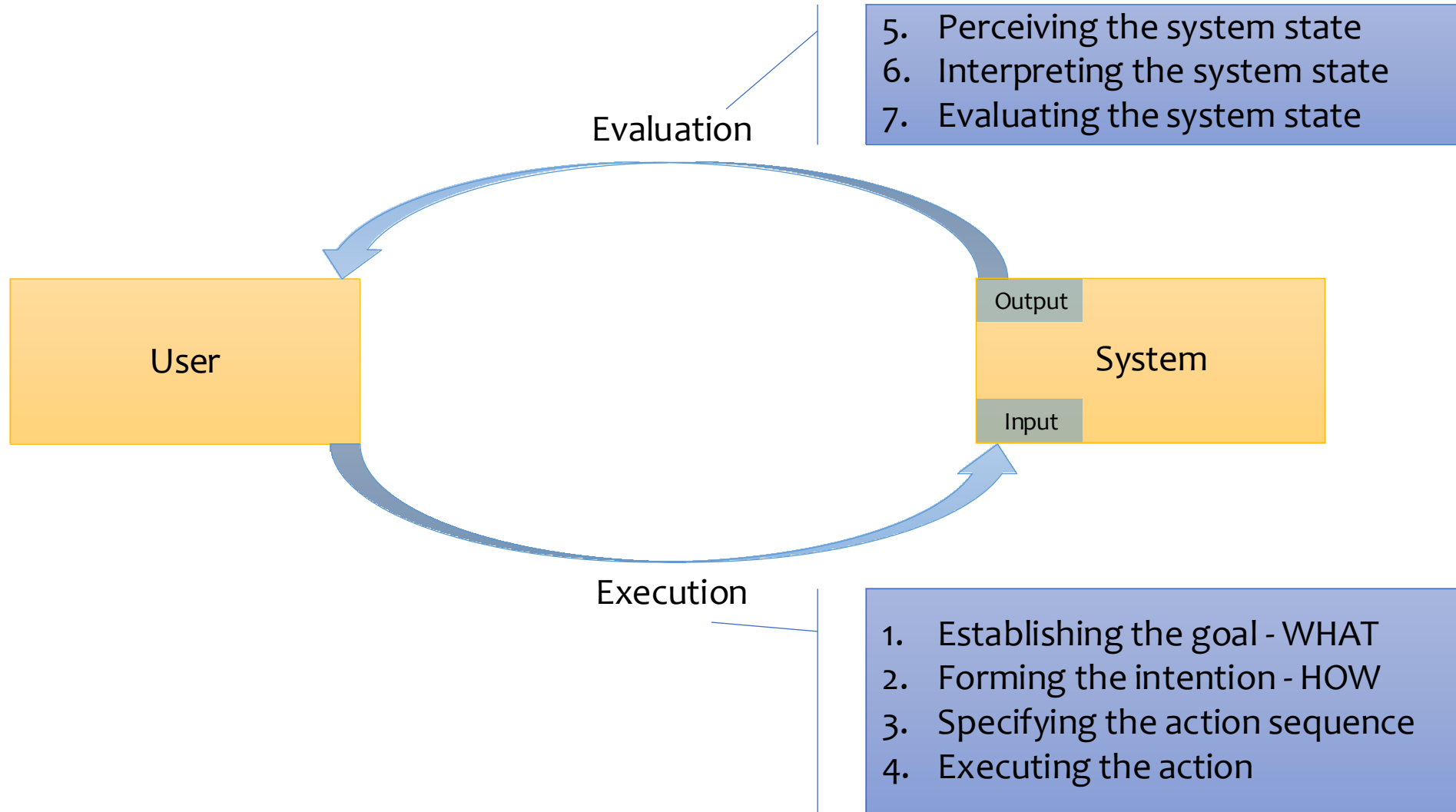


Examples

- Goal: I want to go out of this room



Norman's Model of Interaction





- Goal è il desiderio astratto ad alto livello, indipendente dal sistema informatico. Risponde alla domanda "Cosa voglio ottenere?". Non contiene alcun dettaglio tecnologico. Esempio: "Voglio passare l'esame di HCI".
- Intention è la scelta della strategia funzionale specifica per raggiungere l'obiettivo usando quel determinato sistema. Traduce l'intento in un piano d'azione. Risponde alla domanda "Quale funzionalità o strumento devo usare?" Esempio: "Devo iscrivermi all'appello di HCI tramite la piattaforma AlmaEsami".
- Tasks sequence è l'insieme di operazioni fisiche e cognitive dettagliate necessarie per attuare l'intenzione. Risponde alla domanda "Quali azioni devo fare e in che ordine?" Esempio: 1. Aprire il browser; 2. Andare su AlmaEsami; 3. Fare il login con le credenziali; 4. Cliccare sul pulsante "Iscriviti"

Part of Needfinding (Questions with Goal perspective)

Design Implications

Il progettista deve porsi queste domande per assicurarsi che l'interfaccia guidi l'utente senza generare ostacoli cognitivi o errori.

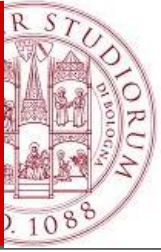
Related to the actions that the designer have to do on the system

Stage of Action		Design Question: How easily can the user ... ?
1	Form the goal	Determine the function of the product or system
2	Form the intention	Tell what actions are possible
3	Specify an action	Determine mapping from intention to action
4	Execute the action	Perform the action
5	Perceive the state of the world	Tell if the system is in the desired state
6	Interpret the state of the world	Determine the mapping from system state to interpretation
7	Evaluate the outcome	Tell what state the system is in

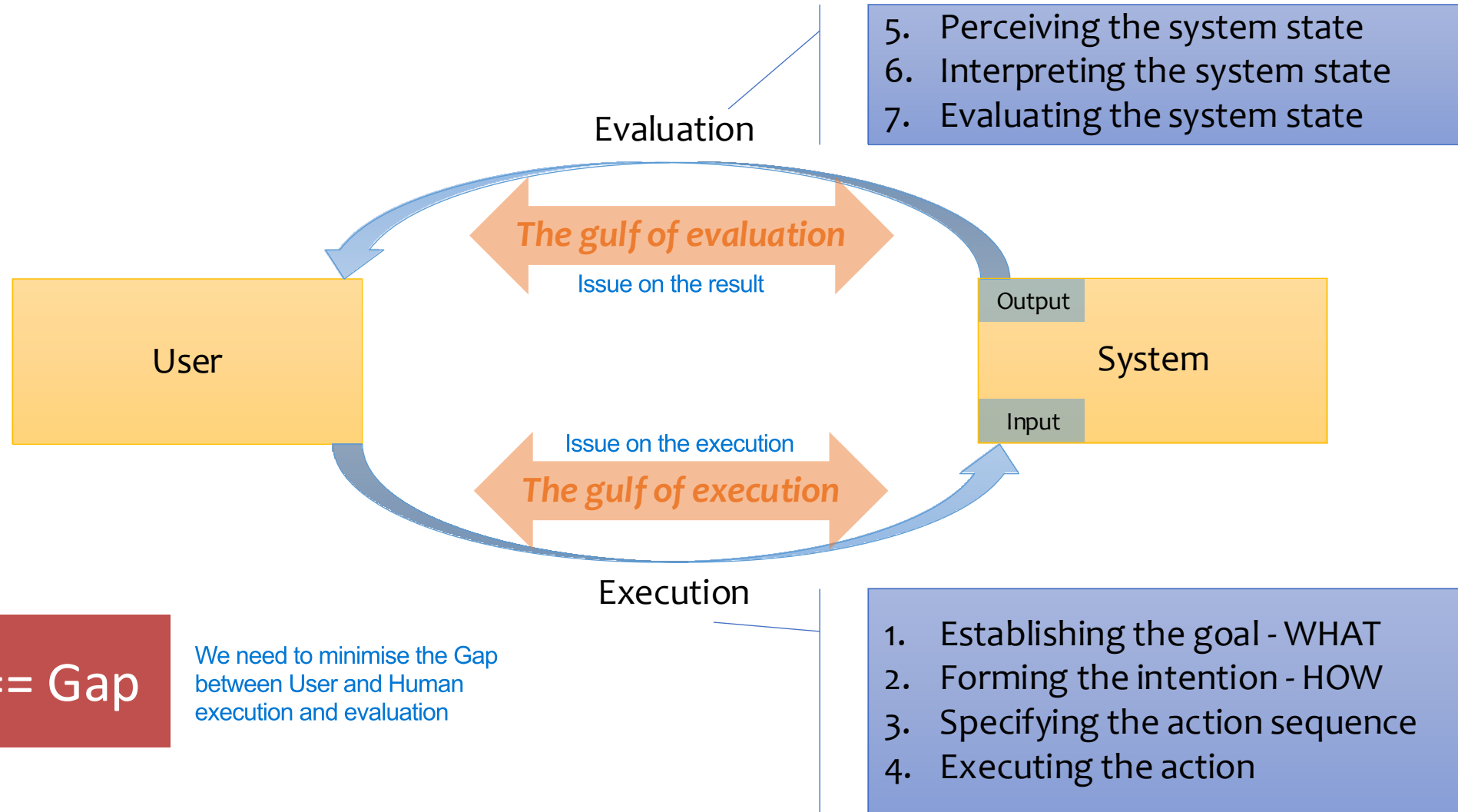
<https://learnline.cdu.edu.au/units/hit381/resources/popups/normantheoryofaction.html>

- Doors...





Norman's Model of Interaction





Gulf of execution: is the distance between the user's goals and the means of achieving them through the system

- When problems in the execution phase occur, the "gulf of execution" widens
- Here, the distance between the users' goals and their ability to use the system increases
- Some examples:

- "What do I do now?"
- "How do I [...]?"
- "I wanted to [...], but I can't see how I would do that?"
- "Do I press this?"
- "Do I type this in here?"
- "I'm looking for [...] but I can't find it"

I stop before doing something on the system

Gulf of evaluation: the amount of effort required to determine the system state

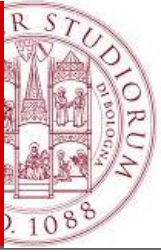
Did something, but the result is not what the user expect

- When problems occur in the evaluation phase, the "gulf of execution" widens
- Here, the distance between the system response and the users' goals widens
- Some examples

- "What happened then?"
- "What did I do wrong"
- User repeats the action (e.g. clicks the NEXT or SUBMIT button)

Come ridurlo: Progettando interfacce con buona affordance (oggetti che suggeriscono visivamente il loro uso), comandi ben visibili e mappature intuitive tra l'azione e l'effetto desiderato.

Come ridurlo: Fornendo feedback immediati e trasparenti (indicatori di caricamento, cambi di colore, suoni o messaggi di stato chiari) che permettano all'utente di capire subito in che stato si trova il sistema.



Norman's Model of Interaction

- We want to minimize the “distance” between the user and the system, so minimize the
 - Gulf of execution
 - Gulf of evaluation
 - Back to our example... Doors!!
 - Affordance!!
 - “a hint provided by an object’s shape and position is called an **affordance**.”
- Una "Norman Door" (Porta di Norman) è una porta il cui design rende poco chiaro se debba essere spinta o tirata per essere aperta.
- A **Norman Door** is a door that is unclear if you push or pull — primarily because of the design prioritizing aesthetic over usability; **form over function**.





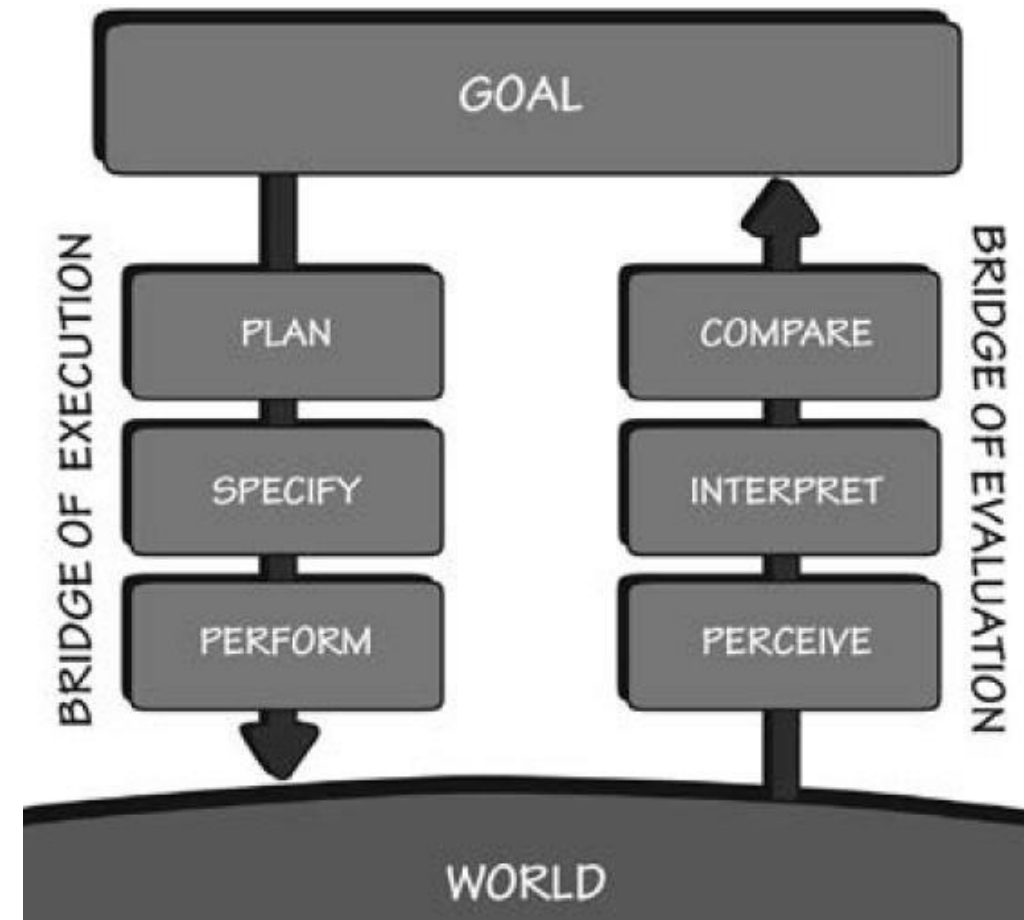
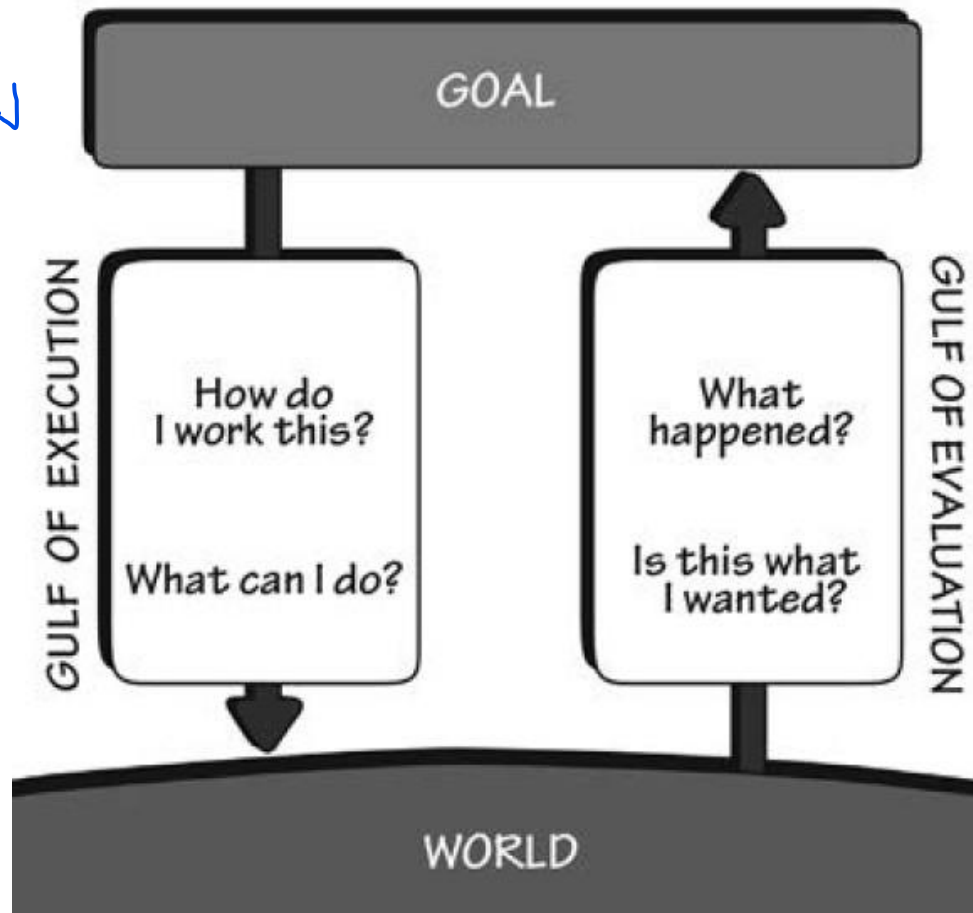
Everything is described from the pov of the user (the computer is a tool to achieve the goal)

Norman's Diagrams

The important point here is that you need to minimise the distance between User and System

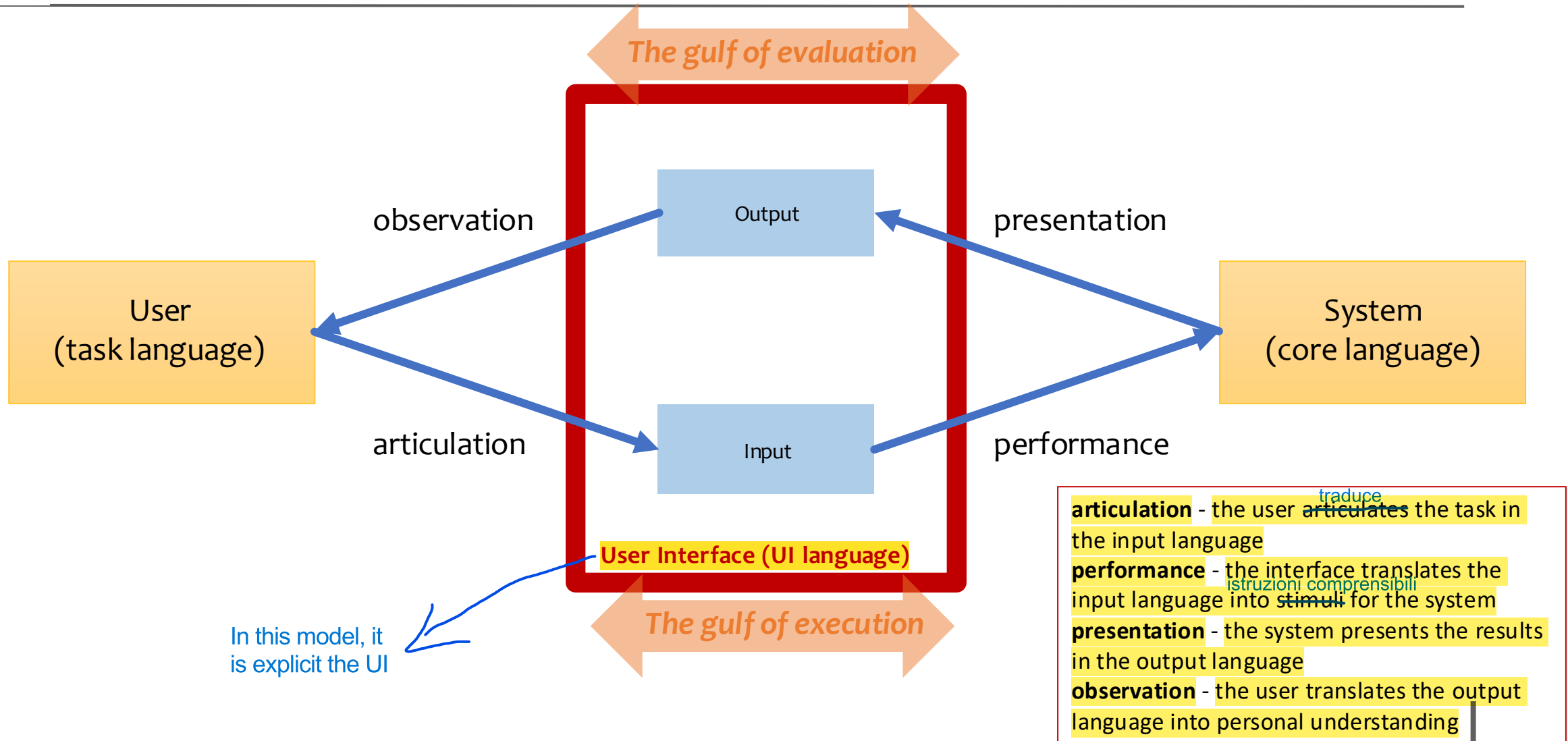
1. **Goal** (form the goal)
2. **Plan** (the action)
3. **Specify** (an action sequence)
4. **Perform** (the action sequence)
5. **Perceive** (the state of the world)
6. **Interpret** (the perception)
7. **Compare** (the outcome with the goal)

This is another way to see the Norman Model of Interaction (with diagram)





Abowd and Beale Model, with Explicit UI





Two type of Human Errors (is never Human error, but system error)

Human Errors* (system errors?!) in the gulf of execution

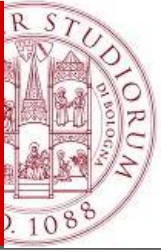
Evaluation of the system

- **Slip**

- ^{User} You have formulated the right action, but fail to execute that action correctly
 - E.g., click the wrong icon, or double-click too slow, ...
- ^{How to solve?} May be corrected by a better interface (spacing, layout, highlights, ...)

- **Mistake**

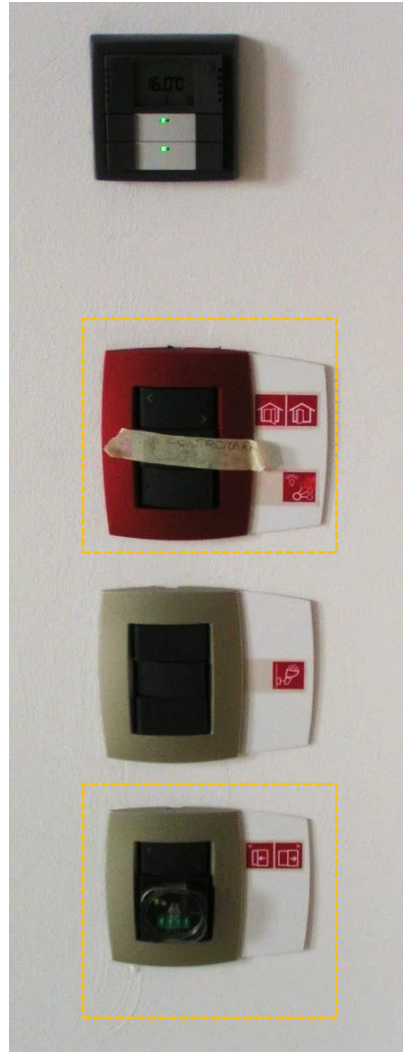
- ^{User} You don't know the system well and you may not formulate the right goal
 - E.g., click  for Zoom, but it means Search ^{Interface error}
- The user's mental model of the system's state is not correct
 - ^{How to solve?}
- Requires more radical redesign, or additional training



* About Human Errors

- Human errors should **never** be considered as faults of the user
- Rather, «they are usually a result of bad design» (Norman) not because of Human faults
- Humans tend to be imprecise, distracted, not-omniscient
 - System design should anticipate this human behavior
 - Minimize the chance of inappropriate actions (evaluation)
 - Maximize the possibility of discovering and repairing an inappropriate action (execution)
 - Enable users to understand the state of the system and build an appropriate model

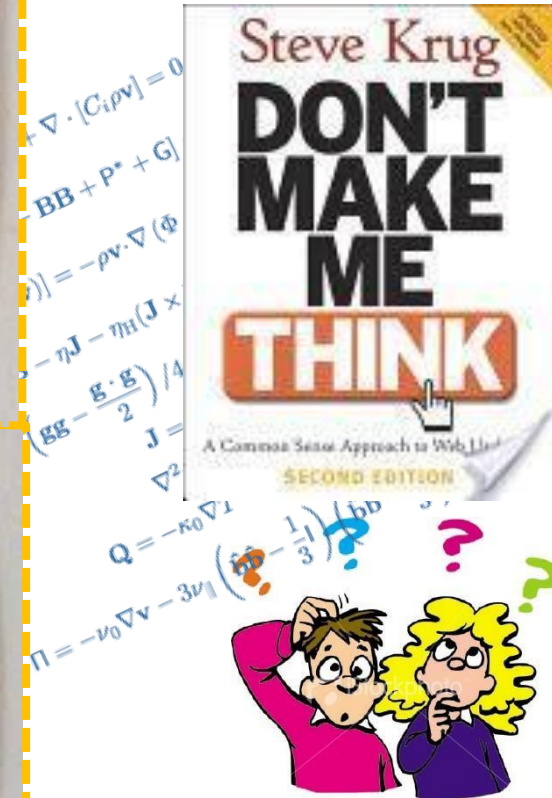
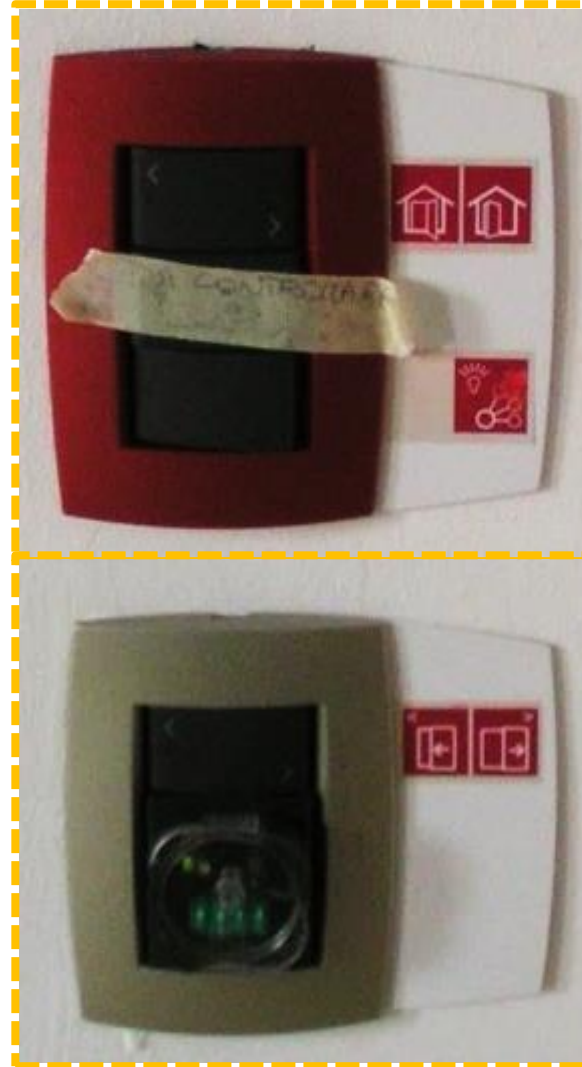
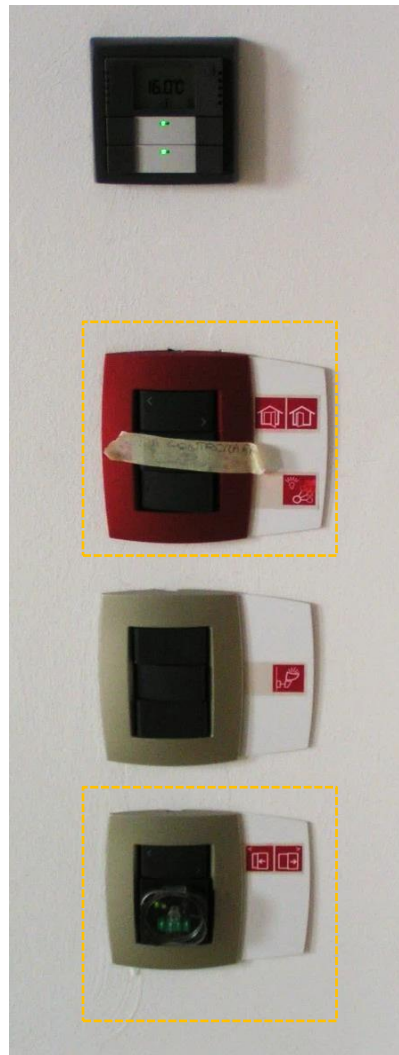
Example (articulation): find the right switch



$$\begin{aligned} \frac{\partial(C_i \rho)}{\partial t} + \nabla \cdot [C_i \rho \mathbf{v}] &= 0 \\ \frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot [\rho \mathbf{v} \mathbf{v} - \mathbf{B} \mathbf{B} + \mathbf{P}^* + \mathbf{G}] &= -\rho \nabla \phi - \nabla \cdot \Pi \\ \frac{\partial E}{\partial t} + \nabla \cdot [(E + P^*) \mathbf{v} - \mathbf{B}(\mathbf{B} \cdot \mathbf{v})] &= -\rho \mathbf{v} \cdot \nabla(\phi + \phi) + \rho H - \rho^2 \Lambda - \nabla \cdot \mathbf{Q} - \nabla \cdot (\Pi \cdot \mathbf{v}) \\ \frac{\partial \mathbf{B}}{\partial t} - \nabla \times [\mathbf{v} \times \mathbf{B} - \eta \mathbf{J} - \eta \Pi(\mathbf{J} \times \mathbf{B}) - \eta \Lambda \mathbf{D}(\mathbf{J} \times \mathbf{B}) \times \mathbf{B}] &= 0 \\ \mathbf{g} &= -\nabla \phi \\ \mathbf{G} &= \left(\mathbf{g} \mathbf{g} - \frac{\mathbf{g} \cdot \mathbf{g}}{2} \right) / 4\pi G \\ \mathbf{J} &= \nabla \times \mathbf{B} \\ \nabla^2 \phi &= 4\pi G \rho \\ \mathbf{Q} &= -\kappa_0 \nabla T - \kappa_{\parallel} \hat{\mathbf{b}} \hat{\mathbf{b}} \cdot \nabla T \\ \Pi &= -\nu_0 \nabla \mathbf{v} - 3\nu_{\parallel} \left(\frac{\partial \mathbf{v}}{\partial t} - \frac{1}{3} \nabla \cdot \mathbf{v} \right) \left(\hat{\mathbf{b}} \hat{\mathbf{b}} - \frac{1}{3} \mathbf{I} \right) : \nabla \mathbf{v} \end{aligned}$$

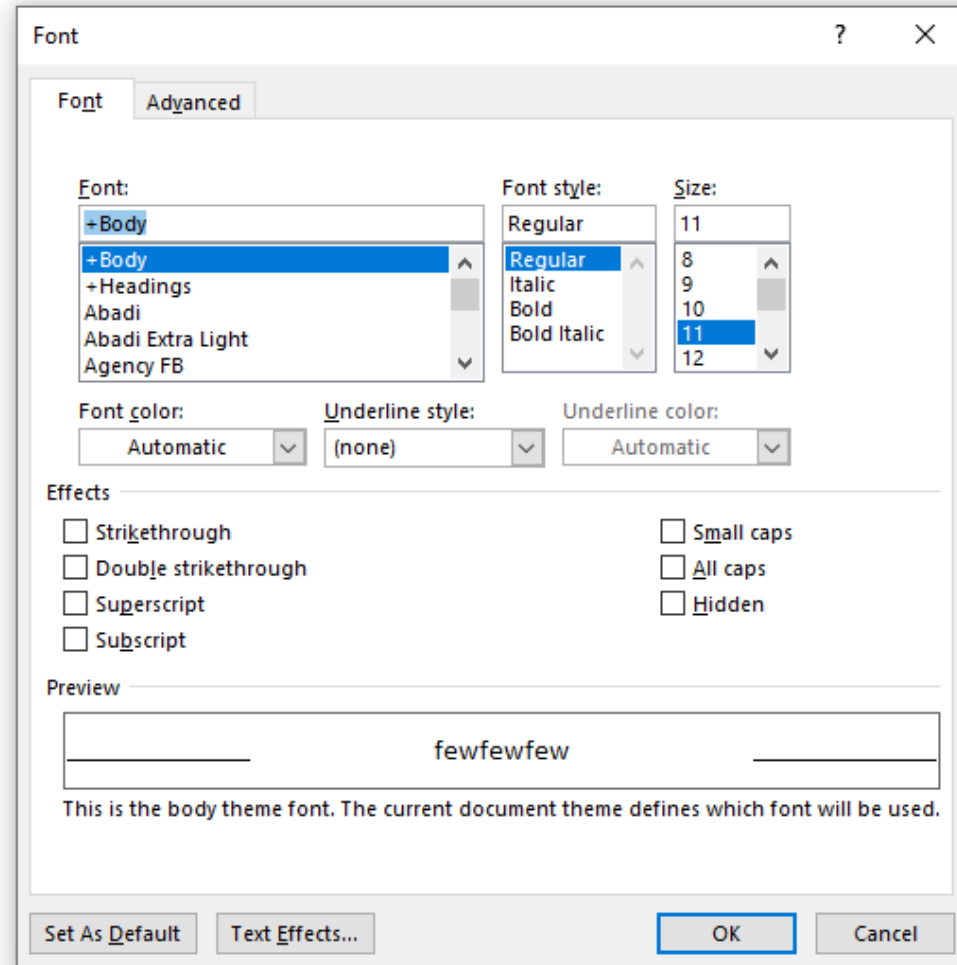


Example (articulation): find the right switch



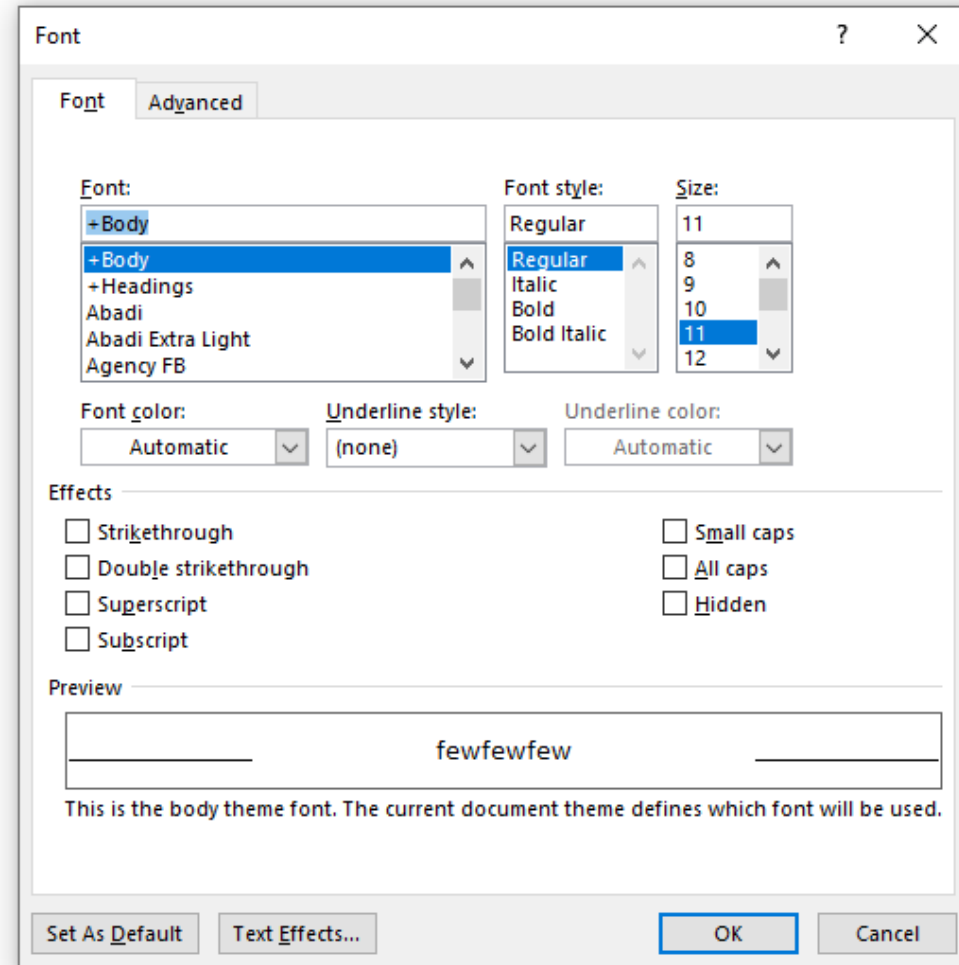


Example (presentation): Which are the allowed combinations?





Example (presentation): Which are the allowed combinations?



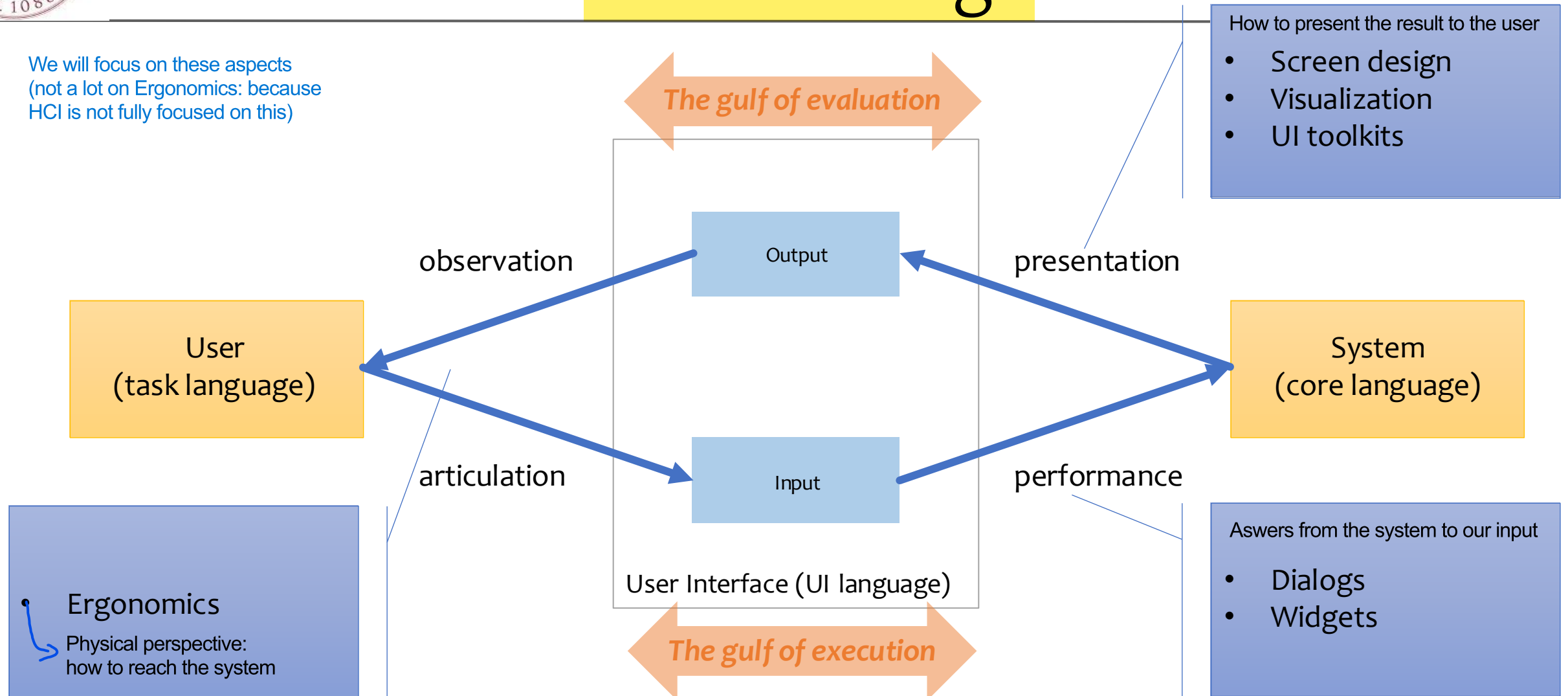
Checkboxes!!

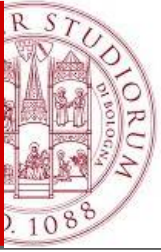
The logic is
correct, the
presentation no!!



Tools, Techniques and Environments for HCI design

We will focus on these aspects
(not a lot on Ergonomics: because
HCI is not fully focused on this)





not to focus too much on these (not all)

Frameworks: Major UI Styles

- Command line interface
- Menus
- Natural language
- Question/answer and query dialog
- Form-fills and spreadsheets
- Windows, Icons, Menus and Pointers (WIMP)
- Mobile
- Point and click
- Three-dimensional interfaces

Design Processes and Frameworks

Approaches for shaping the design process



1

You only ask the User at the start and at the end of the Design process

Before starting the Design of something, we are going to ask Users (we start from Users)

One method of Design process (we will focus only on this)

User-Centered Design (UCD)

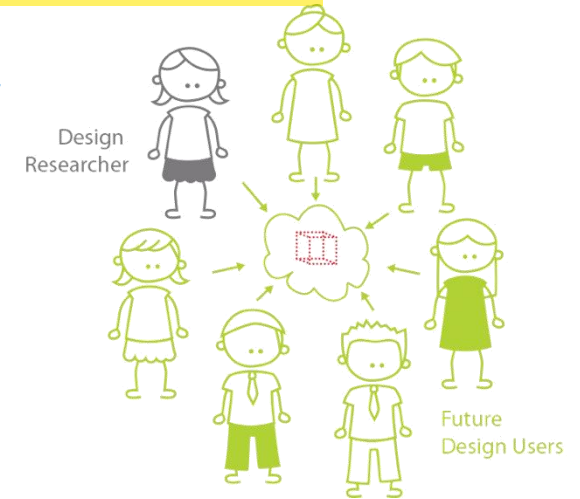
- Avoid the risk of software project failure
 - Estimated 50% are affected by bad developer<->user/client communication
- UCD takes the needs, wants, and limitations of the **actual end users** into account
- **during each phase** of the design process
 - User-centered design issues are discovered during the early stages
- Benefits: systems easier to learn, with faster performance, with less human errors, encourage users to discover advanced features, and avoids “building the wrong system”
- Issues: how to find users? How many? How motivated? How to speak their language?
- How to extract user needs, business needs, organizational implications?

Participatory Design

Co-design with Users
(but takes a lot of time)

- One step further than UCD, users are directly involved in the collaborative design of the things and applications they use

All decision, before taking them, are asked to the Users
- Engage a group of users
 - Discussions
 - Creating scenarios, sketches, dramatizations
 - Creating and testing lo-fi prototypes
 - Continuous meetings, flexible management
 - Highly reliant on the skills of the group moderators/leaders (keep involved, filter ideas, reward participation, work around resistances, ...)
 - More effective with more mature and prepared user populations (less with kids, elderly, disabled, ...)



Agile Interaction Design

- Borrows ideas from Agile development in software engineering
- Key: evolutionary development
 - System is built incrementally in rapid release cycles
 - Rapid prototyping techniques (for hardware, software, and physical objects)
- Focus on low-cost many-iterations prototypes
- Requires fast usability inspection (extreme usability, XU)
- Makers' culture (only if it involves users!)

Design Thinking

Il Design Thinking è una metodologia di risoluzione dei problemi incentrata sull'essere umano (human-centered), nata per guidare l'innovazione bilanciando tre elementi chiave:

- Umani: Soddisfare i reali bisogni delle persone.
- Tecnologia: Sfruttare le possibilità offerte dalle tecnologie esistenti o emergenti.
- Business: Rispettare i requisiti di sostenibilità economica e successo aziendale.

- “A human-centered approach to innovation that draws from the designer’s toolkit to integrate the needs of people, the possibilities of technology, and the requirements for business success.”

si può tornare indietro in qualsiasi momento in base a ciò che si impara dai test.

— [Tim Brown, IDEO’s Executive Chair](#)

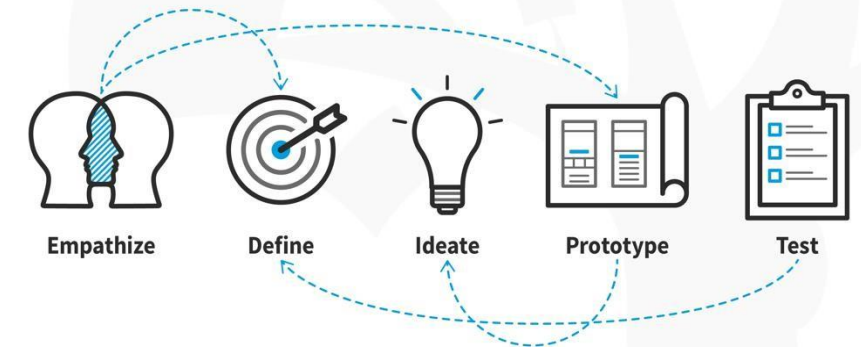
- A 5-stage, non-linear and iterative, process
1. **Empathize** – research users’ needs
 2. **Define** – state the found needs
 3. **Ideate** – challenge assumptions and ideate
 4. **Prototype** – create solutions
 5. **Test** – try the solutions out

Trasformare le idee più promettenti in soluzioni fisiche o digitali concrete ma economiche e veloci da realizzare

Mettere i prototipi nelle mani degli utenti per osservare come li usano e raccogliere feedback

Mettere in discussione le supposizioni di partenza e generare il maggior numero possibile di idee e soluzioni creative attraverso sessioni di brainstorming.

Design Thinking: A 5-Stage Process



Interaction Design Foundation
interaction-design.org



Service Design

Related to design itself

- Describe the contemporary shift from *products* (e.g., a car of a specific brand) to *services*, e.g., the car as a tool for an elderly customer that wants to take an Uber ride to visit a friend
- Focus on the complete experience, including business resources and processes
- Build upon five key principles, according to “This is Service Design Thinking”:
 1. User-centered – focus on all users needs
 2. Co-creative – include all relevant stakeholders Coinvolge attivamente nel processo di progettazione tutti gli stakeholder rilevanti
 3. Sequencing – break a complex service into separate processes
 4. Evidencing – envision service, not product, experiences Progetta ed evidenzia gli elementi tangibili o visivi che rendono concreta un'esperienza immateriale
 5. Holistic – design across networks of users and interactions
↳ Considera l'intero contesto dell'esperienza, progettando l'intera rete di utenti, interazioni, risorse aziendali e processi organizzativi sottostanti.

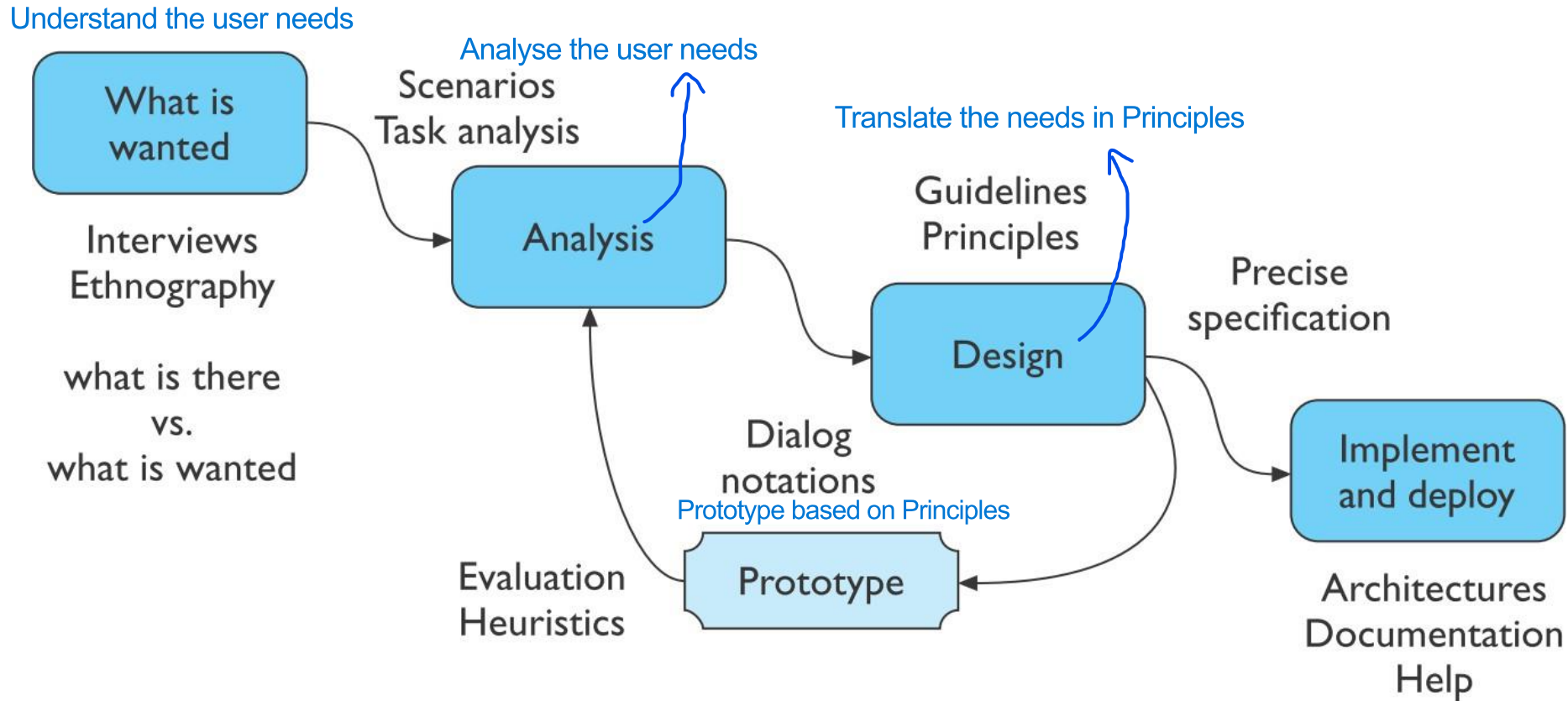
Human-centered Design Process

A process for designing interactive systems with a focus on usability



Human-Centered Design Process

(simplified and generic)



Going in the loop (Analysis-Design-Prototype) until we are happy about the prototype (that will be the final one)



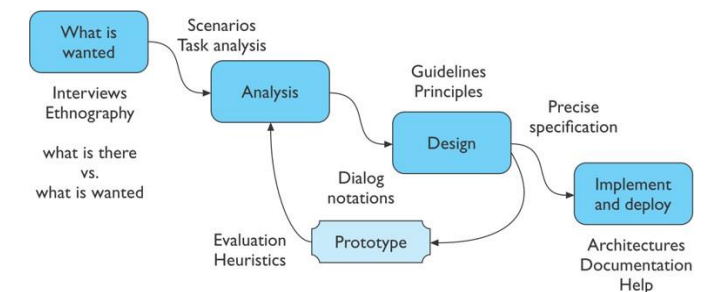
Human-Centered Design Process – The Main Steps (1)

- **Needfinding** – what is needed

- What exactly is needed? How are people currently accomplishing the goal?
- User observation, interviews, ...

- **Analysis**

- Formalize and structure the needs
- Create interaction scenarios, stories, tasks
- Compare current situation with expected new situation

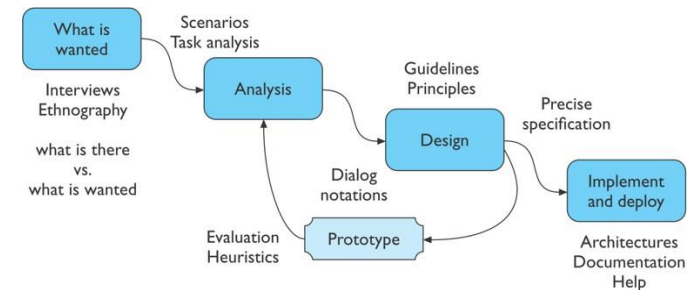


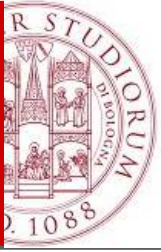


Human-Centered Design Process – The Main Steps (2)

■ Design

- The main choices to shape the system
- Rules, guidelines, design principles
- Considering different types of users
- Modeling and describing interaction
- Visual layout
- Consider all inputs from cognitive models, communications theories, organization issues

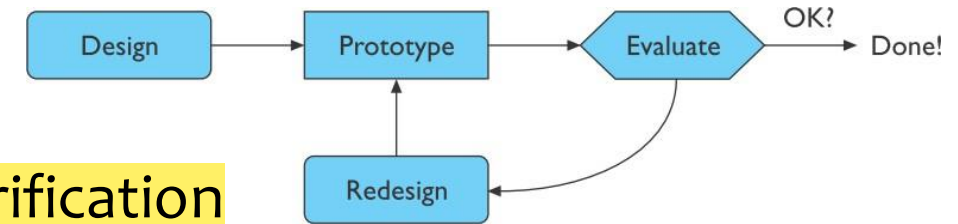




Human-Centered Design Process – The Main Steps (3)

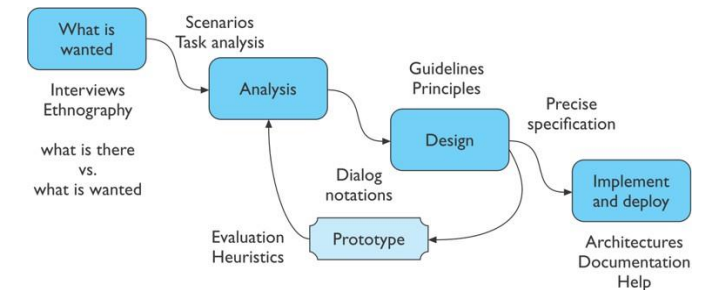
■ Iteration and prototyping

- Design must be supported by intermediate verification
- Evaluate the design in its partial forms:
 - Prototypes
 - Evaluation metrics
- Involving users



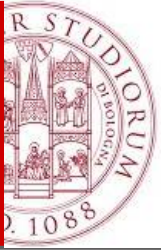
■ Implementation and deployment

- Hardware and software implementation
- Documentation



Usability (ISO standard definition)

- **Usability:** “extent to which a system, product or service ^{is usable if it} can be used by specified users to achieve specified goals with *effectiveness, efficiency* and *satisfaction* in a specified context of use”
 - Note 1: The “specified” users, goals and context of use refer to the particular combination of users, goals and context of use for which usability is being considered
 - Note 2: The word “usability” is also used as a qualifier to refer to the design knowledge, competencies, activities and design attributes that contribute to usability, such as usability expertise, usability engineering, usability testing, etc.



Usability

- **Usability:** how well users can use the system's functionality
- **Dimensions of usability:**
 - **Usefulness:** does it do something people want?
 - **Learnability:** is it easy to learn?
 - **Memorability:** one learned, is it easy to remember?
 - **Effectiveness:** does it allow reaching the goal?
 - **Efficiency:** once learned, is it fast to use?
 - **Visibility:** is the state of the system visible?
 - **Errors:** are errors few and recoverable?
 - **Satisfaction:** is it enjoyable to use?

Usability: Don't Make Me Think



OK. This looks like the product categories...

Laptops, Memory... There it is: Monitors. *Click*

...and these are today's special deals.



Hmm. Pretty busy. Where should I start?

Hmm. Why did they call it that?

Can I click on that?

Is that the navigation? Or is that it over there?

Why did they put that there?

Those two links seem like they're the same thing. Are they really?

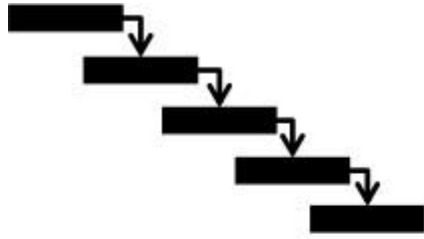
You can put HCI when ever you need it (when
there is interaction between Human and System)

HCI in the Software Process

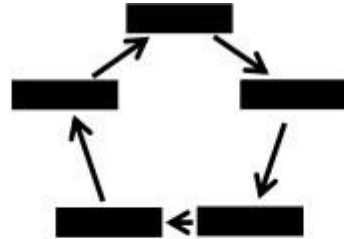
How to include Human-centered processes in Software Engineering

Software Engineering Processes

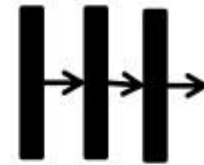
Where / how does HCI fit in?



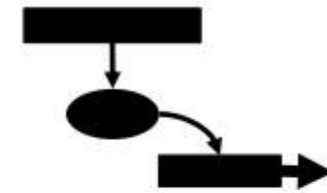
Waterfall



Iterative
waterfall



Agile
(scrum)

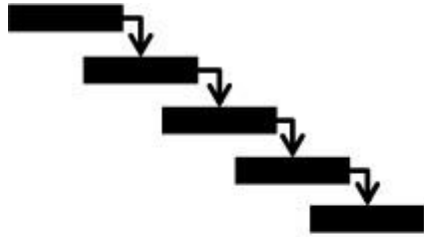


Lean

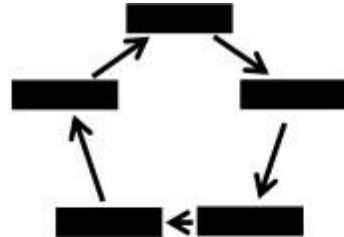
Software Engineering Processes

Where / how does HCI fit in?

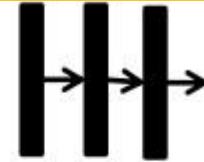
Always a step ahead!



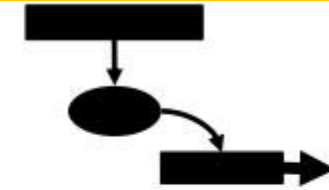
Waterfall



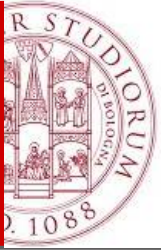
Iterative
waterfall



Agile
(scrum)



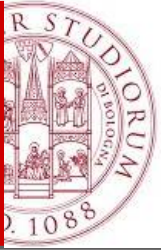
Lean



Always a Step Ahead (I)

Can be put:

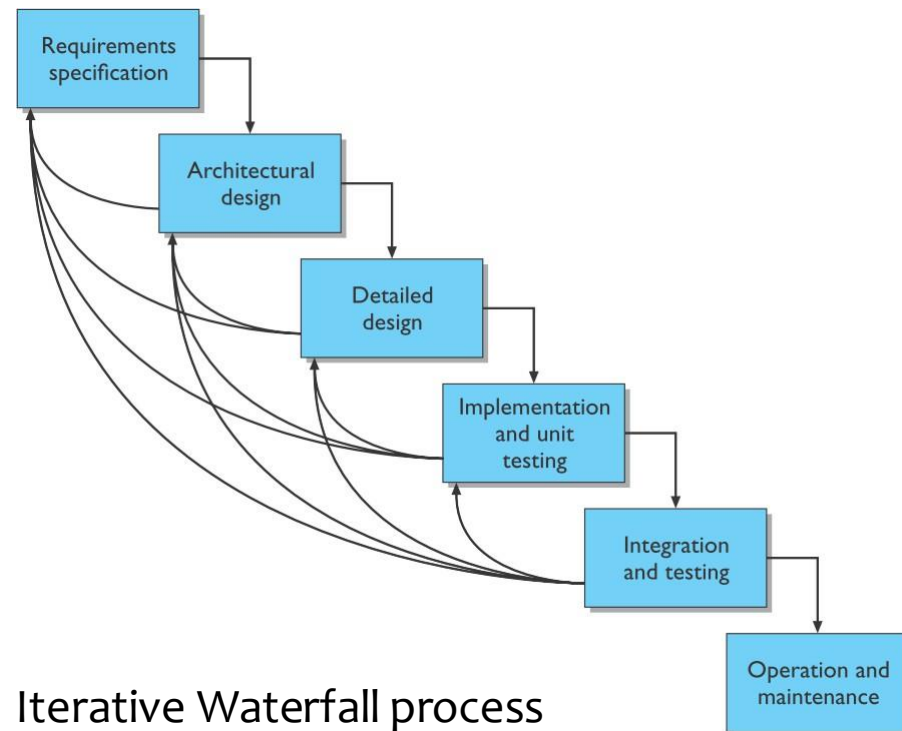
- Before
 - Every design step
 - Every implementation step
 - Any product iteration (or sprint)
 - ...
- You need a user-centered step
 - Evaluate usability
 - Experiment with users
 - Evaluate alternative flows
 - Evaluate alternative layouts
 - ...



Always a Step Ahead (II)

- Usability, Safety, Performance, are part of Non-Functional Requirements
- User-centered steps are cheaper than development
 - User research about users' needs to decide what to design
 - Heuristic evaluations before testing with users
 - Evaluating prototypes instead of full-fledged products
- Anticipate critical decision points later in the project

Example



- Each step must be
 - Preceded with user evaluation of the design choices and formalized requirements
 - Followed with user evaluation of the result
- Must produce additional artifacts to allow this kind of iteration
 - Prototypes

References

- Robert Miller, MIT Course “6.813/6.831: User Interface Design & Implementation”
 - Spring 2018 - <http://web.mit.edu/6.813/www/sp18/>
 - Spring 2011 - <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-831-user-interface-design-and-implementation-spring-2011/index.htm>
- Dix *et al*: Human Computer Interaction
 - Chapters 3, 5
- Norman: The Design of Everyday Things
 - Chapter 2
- Krug: Don't make me think
 - Introduction
- Shneiderman: Designing the User Interface
 - Chapter 4

These slides are a modified version of the ones of prof. Luigi De Russis, PoliTo. A big thanks to Luigi De Russis (PoliTo) for his work!